

Critical Orbit Tiling and an All-Band Large-Sieve Reduction for the Actual-Coefficient Residual Determinant Family

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Abstract

The preceding twin-prime-correlation (TPC) analysis transferred a mesoscopic shift-spectral gap to one canonical orbit-frequency band of the actual residual coefficient field, but left the orthogonal complement uncontrolled. We resolve the resulting all-band geometry. On the quotient circle associated with a fixed progression modulo H_0 , we construct an exact partition into L translated residue-comb cells and the corresponding orthogonal orbit projectors. Every determinant modulation $m\alpha$ then factors through an exact input-output band matrix whose nonzero entries are resonance tubes. At the critical resolution $L \asymp J$, every band has bounded physical-window trace and each tube has α -half-width $(JM)^{-1} = X^{-1+o(1)}$. Thus the original fine-frequency scale is restored as a complete square-function decomposition, with no band cover-count loss.

All-band Parseval alone is not a large-sieve estimate. We prove the sharp boundary: for arbitrary independent orbit inputs on K source slopes, the unlabelled determinant synthesis has operator norm squared exactly K . The actual projective layers possess additional structure, namely one common smooth orbit profile. Exploiting it, we prove the Hilbert-valued bound

$$\|\widehat{\mathcal{F}}^\omega(\alpha)\|^2 \ll_{\phi_\omega, H_0} \frac{J}{H_0} \min\left\{K, 1 + \frac{1}{J\|\alpha\|_{\mathbb{T}}}\right\} \sum_m \|B_{\omega, m}(\alpha)\|^2$$

for $\alpha \neq 0$, under an explicit no-wrap condition; at $\alpha = 0$ the collision factor is K . The factor is sharp, up to constants, for common-profile slope boxes containing the corresponding consecutive cluster. Minkowski reassembles the frozen family built from the complete signed actual h_0 -coefficient field and produces a canonical projective m -square envelope; at each fixed m , distinct cofactors retain their literal residual labels $d(m)r$ and are orthogonal before residual grouping.

For $M = X^{267/400+o(1)}$ and $J = X^{133/400+o(1)}$, a dense slope box has critical coarse-edge collision exponent $67/200$. In that sharp model class the common profile saves the full orbit length over arbitrary-slope synthesis, but its loss still exceeds the TPC allowance $X^{1/400+o(1)}$ by $X^{133/400+o(1)}$. No matching lower bound is assumed for the literal slope support. We also prove exact first and second moments for the two-dimensional tube load and a logarithmic annular resonance budget relative to any fixed representation with bounded displayed projective square envelope. Consequently the orbit-band reassembly problem is reduced, not the fixed-shift prime-pair problem: atomic normalization, internal prime cancellation, residual grouping, alias unfolding, and localization of the full physical atom remain open. No parity-barrier breach or twin-prime conclusion is claimed.

Keywords: determinant equation; orbit-frequency tiling; resonance tubes; vector-valued large sieve; Möbius correlation; square functions; parity problem.

Convention. We write $e(x) = \exp(2\pi i x)$, identify $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and put $\|\alpha\|_{\mathbb{T}} = \min_{k \in \mathbb{Z}} |\alpha - k|$. Fourier transforms on $\mathbb{R}$ use $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e(-x\xi) dx$; the unitary Fourier transform on $\ell^2(\mathbb{Z})$ is $(\mathcal{F}y)(\beta) = \sum_{j \in \mathbb{Z}} y_j e(-j\beta)$.

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1 Introduction

The large-prime descent in the TPC sequence produces the determinant equation

$$pr - mj = h_0, \quad (1)$$

with $m \asymp M$, an orbit length J , and a literal residual coordinate $s = d(m)r$. TPC-46 completed the consecutive-shift family and found a mesoscopic shift-spectral gap when the output vector is independent of the orbit coordinate j [6]. TPC-47 then projected the actual j -dependent residual coefficient onto one complete residue-comb band. Its canonical low core vanishes there on a nonempty shift interval, while the full frozen family has arbitrary-power leakage [7]. That theorem is genuinely sectorial: it neither controls the orthogonal orbit-frequency complement nor gives a lower bound for the mass retained by the chosen band.

The natural continuation is to tile the entire orbit-frequency circle and retain orthogonality through determinant dilation. Two different questions are hidden in that sentence.

- (i) Can the full orbit field be decomposed into exact input-output resonance channels without paying the number of bands?
- (ii) Does that decomposition by itself reduce coherent summation over the source slopes m ?

The answer to the first question is yes. The answer to the second is no for arbitrary orbit inputs, but becomes quantitatively positive for the common smooth profiles supplied by the actual projective reassembly. This positive/negative distinction is the main result of the paper.

1.1 Critical orbit tiling

Let $G = H_0^{-1}\mathbb{Z}/\mathbb{Z} \subset \mathbb{T}$, and partition the quotient $\mathbb{T}/G$ into L half-open cells. Their inverse images $\mathcal{C}_k$, $k \in \mathbb{Z}/L\mathbb{Z}$, form translated full residue combs of measure $1/L$. If Π_k is the corresponding orbit Fourier projector, then

$$\Pi_k \Pi_\ell = 0 \quad (k \neq \ell), \quad \sum_{k \bmod L} \Pi_k = I \quad (2)$$

in the strong operator topology. On a scalar orbit window of N integers,

$$\mathrm{tr}(E_{I,v}^* \Pi_k E_{I,v}) = \frac{N}{L}. \quad (3)$$

Here $E_{I,v}$ denotes zero extension on one fixed unit non-orbit fiber; it is defined precisely in [section 3](#). The critical choice $L \asymp J$ therefore has bounded trace per band. It sacrifices the growing single-band trace of TPC-47, but no mass hypothesis is then needed because the orthogonal sum of all bands is the entire orbit space.

For the modulation

$$(U_{m,\alpha}y)_j = e(mj\alpha)y_j, \quad (4)$$

the block $\Pi_k U_{m,\alpha} \Pi_\ell$ is supported precisely where the shifted input cell meets the output cell. After setting $q = k - \ell \pmod{L}$, the support condition is

$$d_G(m\alpha, c_q) < \Delta, \quad c_q = \frac{q}{LH_0}, \quad \Delta = \frac{1}{LH_0}. \quad (5)$$

Here $d_G(x, y) := \text{dist}(x - y, G)$. Thus every channel is a union of exact determinant resonance tubes. At $m \asymp M$, their half-width in α is

$$\frac{\Delta}{m} \asymp \frac{1}{LM}. \quad (6)$$

The critical choice $L \asymp J$ restores the fine scale $(JM)^{-1} = X^{-1+o(1)}$. This is a square-function resolution, not a spectral gap.

1.2 The arbitrary-input barrier

Write $K = |\mathcal{M}|$ for the number of supported slopes and define the unlabelled synthesis

$$T_\alpha((y_m)_{m \in \mathcal{M}}) := \sum_{m \in \mathcal{M}} U_{m,\alpha} y_m. \quad (7)$$

As an operator from $\bigoplus_m \tilde{\mathcal{H}}$ to $\tilde{\mathcal{H}}$, it satisfies the exact identity

$$\boxed{\|T_\alpha\|^2 = K}. \quad (8)$$

Indeed, the upper bound is Cauchy–Schwarz, while $y_m = U_{m,\alpha}^{-1} g$ makes every output equal to the same vector g . Decomposing each y_m into all input bands does not change this example: the input band can compensate the determinant translation. Consequently all-band Parseval is not, by itself, a large-sieve gain.

We make the obstruction more precise by retaining the difference channel q . The squared norm of the resulting channel-lift operator is exactly the essential maximum $R_L(\alpha)$ of the pointwise resonance-tube multiplicity. The physical channel collapse has norm squared L , and

$$K \leq LR_L(\alpha). \quad (9)$$

Both constants are optimal in their declared arbitrary-input classes. No statement here says that the actual TPC coefficient saturates these classes.

1.3 The common-profile large sieve

The inherited projective representation has additional layerwise structure. Every projective layer has one common sampled profile $\phi_\omega(j/J)$ for all source slopes. After grouping the prime and cofactor variables at fixed m , its full shift transform is

$$\hat{\mathcal{F}}^\omega(\alpha) = \sum_{\substack{j \in \mathbb{Z} \\ j \equiv a_\omega \pmod{H_0}}} \phi_\omega(j/J) \left(\sum_m B_{\omega,m}(\alpha) e(mj\alpha) \right) \otimes e_j. \quad (10)$$

Let $t = \|\alpha\|_{\mathbb{T}}$, with $D_J(0; K) := K$. Under the no-wrap condition

$$H_0(M_+ - M_-)t \leq \frac{1}{4}, \quad (11)$$

we prove

$$\boxed{\|\hat{\mathcal{F}}^\omega(\alpha)\|^2 \ll_{\phi_\omega, H_0} \frac{J}{H_0} D_J(t; K) \sum_m \|B_{\omega,m}(\alpha)\|^2}, \quad D_J(t; K) := \min \left\{ K, 1 + \frac{1}{Jt} \right\} \quad (t > 0). \quad (12)$$

The proof is a progression Poisson estimate followed by the Schur test for the vector-valued Gram matrix, in the classical large-sieve tradition [1]. A cluster of consecutive slopes shows that

$D_J(t; K)$ is sharp up to constants for common-profile slope boxes containing a sufficiently long consecutive cluster.

The full signed actual coefficient is not one projective layer. Minkowski reassembles all layers and leaves a declared projective m -square envelope. This is an exact reduction, not an estimate of that envelope by the physical atomic diagonal. At fixed m , different cofactors r occupy the distinct residual coordinates $d(m)r$, so the remaining inner object is an exact r -square sum of prime exponential sums.

1.4 Endpoint ledger and claim boundary

At the inherited endpoint,

$$M = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}. \quad (13)$$

The critical band count is $L = X^{133/400+o(1)}$. On a coarse-edge shift scale $t \asymp M^{-1}$, a dense slope box has normalized collision factor

$$D_J(t; K) = X^{67/200+o(1)}. \quad (14)$$

The TPC allowance is only $X^{1/400+o(1)}$, leaving the exact exponent gap

$$\frac{67}{200} - \frac{1}{400} = \frac{133}{400}. \quad (15)$$

Thus, in the dense sharpness class, the large sieve gains the full orbit length relative to arbitrary slopewise Cauchy, but it does not close atomic normalization. This ledger is not a lower bound for the literal slope support, whose cardinality may be smaller.

The analysis concerns the frozen shift family built from the literal physical h_0 -coefficient field. Only $h = h_0$ is the inherited physical residual synthesis, and that object belongs to the low-sign-averaged residual Gram route rather than the native affine physical face. Nothing below proves an atomic Bessel bound, controls the ultra-low or alias complements, reconstructs physical packets at new shifts, breaks the parity barrier, or establishes a prime-pair asymptotic.

2 The inherited actual residual interface

We state the exact coefficient field before introducing any band decomposition. This prevents an abstract all-band identity from being substituted for the physical source-mask coefficient.

2.1 Physical and frozen families

Fix $h_0 \in \mathbb{Z} \setminus \{0\}$, put $H_0 = |h_0|$, and let a source row be $\rho = (\ell_\rho, d_\rho)$, with

$$m_\rho = \ell_\rho d_\rho \asymp M, \quad M = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}. \quad (16)$$

On the inherited dyadic row cell, write

$$0 < M_- \leq m_\rho \leq M_+, \quad M_\pm = X^{267/400+o(1)}. \quad (17)$$

As an inherited support hypothesis from the row parametrization in TPC-45/47, the integer m_ρ determines the row ρ on this cell [5, 7]. Let

$$\mathcal{M} := \{m_\rho : \rho \text{ occurs on the inherited support}\}, \quad K := |\mathcal{M}| \leq MX^{o(1)}. \quad (18)$$

No matching lower bound for the actual value of K is assumed.

Let $\mathcal{O}_L$ be the left-side opposite-row set, and let $\mathcal{S}$ contain all supported residual labels. Put

$$\mathcal{V}_L = \ell^2(\mathcal{O}_L \times \mathcal{S}), \quad \mathcal{H}_L = \mathcal{V}_L \widehat{\otimes} \ell^2(\mathbb{Z}). \quad (19)$$

For a supported cofactor r , the literal physical residual is

$$s = d_\rho r. \quad (20)$$

After removal of zero atoms, d_ρ, r, s are squarefree and $\mu(s) = \mu(d_\rho)\mu(r)$ [4–6].

Let $\gamma_\rho^{(1)}$ be the source coefficient, $\mathfrak{A}_{\rho,\gamma}^{\text{act}}(j)$ the literal joint source-mask multiplier, and $\Lambda(n) = (\log n)\Phi_0(n/X)$. The left physical large-prime residual synthesis is

$$\mathcal{F}_{h_0}^{\text{act},L} = \sum_{\substack{\gamma \in \mathcal{O}_L, j \in \mathbb{Z}, \rho, p, r \\ pr - m_\rho j = h_0}} \gamma_\rho^{(1)} \mathfrak{A}_{\rho,\gamma}^{\text{act}}(j) \mu(r) \Lambda(pr) e_{\gamma, d_\rho r} \otimes e_j, \quad (21)$$

with the inherited prime, terminal, row, and orbit restrictions understood. Every coefficient is zero off those finite supports. TPC–45/46 identify (21) with the negative large-prime residual synthesis $-\sum_p Ux_p$. It has the literal actual coefficients, but is not the native affine physical Gram face; their cross inner products need not agree [5–7].

For every $h \in \mathbb{Z}$, define $\mathcal{F}_h^{\text{fr},L}$ by replacing the determinant equation in (21) with

$$pr - m_\rho j = h \quad (22)$$

and freezing every row, support, coefficient, mask, and residual label constructed at h_0 . Then

$$\mathcal{F}_{h_0}^{\text{fr},L} = \mathcal{F}_{h_0}^{\text{act},L}, \quad \widehat{\mathcal{F}}^{\text{fr},L}(\alpha) = \sum_h \mathcal{F}_h^{\text{fr},L} e(-h\alpha). \quad (23)$$

For $h \neq h_0$, this is an algebraic frozen-coefficient extension, not a newly reconstructed physical TPC packet.

2.2 Coefficientwise signed reassembly

For a supported source integer $m = \ell d(m)$ and cofactor r , define the complete vector profile

$$G_{m,r}^L(j) := \sum_{\gamma \in \mathcal{O}_L} \gamma_m^{(1)} \mathfrak{A}_{m,\gamma}^{\text{act}}(j) e_{\gamma, d(m)r} \in \mathcal{V}_L. \quad (24)$$

Then

$$\mathcal{F}_h^{\text{fr},L} = \sum_{p,r,m,j} \mu(r) \Lambda(pr) G_{m,r}^L(j) \otimes e_j \mathbf{1}_{pr-mj=h}. \quad (25)$$

The TPC–36 projective row-mask theorem, together with the inherited large-prime and Mellin separations, gives an exact signed Bochner representation [2, 3, 7]:

$$\mathcal{F}_h^{\text{fr},L} = \int \mathcal{F}_h^\omega d\nu(\omega), \quad (26)$$

where

$$\mathcal{F}_h^\omega = \sum_{p,r,m} \sum_{\substack{j \in \mathbb{Z} \\ j \equiv a_\omega \pmod{H_0}}} A_\omega(p, r, m) \phi_\omega(j/J) v_{\omega,p,r,m} \otimes e_j \mathbf{1}_{pr-mj=h}. \quad (27)$$

Here $v_{\omega,p,r,m} \in \mathcal{V}_L$ is independent of j and is supported on coordinates $(\gamma, d(m)r)$. Every $\phi_\omega \in C_c^\infty(\mathbb{R})$. After one fixed common cutoff, these profiles have common compact support and uniform fixed-order C^B -seminorm control; for every fixed B , the absolute projective mass weighted by $\|\phi_\omega\|_{C^B}$ is $X^{o(1)}$.

The representation is coefficientwise before terminal summation. In particular, every orthogonal orbit projector below is applied to the already reassembled profile (24); the projective layers are used to prove a common-profile inequality, never to select a favorable summand. Sharp orbit projectors may create sinc tails, so all band decompositions are understood in ℓ^2 , not in ℓ^1 .

3 Critical tiling of the orbit-frequency quotient

Let $\mathcal{V}$ be a Hilbert space carrying every non-orbit label and put

$$\tilde{\mathcal{H}} = \mathcal{V} \hat{\otimes} \ell^2(\mathbb{Z}). \quad (28)$$

The orbit Fourier transform

$$\mathcal{F}_j : \ell^2(\mathbb{Z}) \longrightarrow L^2(\mathbb{T}), \quad (\mathcal{F}_j y)(\beta) = \sum_j y_j e(-j\beta), \quad (29)$$

is unitary.

3.1 Half-open residue-comb cells

Fix integers $H_0, L \geq 1$, let

$$G := H_0^{-1}\mathbb{Z}/\mathbb{Z} \subset \mathbb{T}, \quad \Delta := \frac{1}{LH_0}, \quad c_k := \frac{k}{LH_0} \quad (k \in \mathbb{Z}/L\mathbb{Z}), \quad (30)$$

and use the quotient coordinate $\pi(\beta) = H_0\beta \pmod{1}$. Let

$$I_k := \left[\frac{k-1/2}{L}, \frac{k+1/2}{L} \right) \pmod{1}, \quad \mathcal{C}_k := \pi^{-1}(I_k). \quad (31)$$

Thus $\mathcal{C}_k$ is the union of H_0 half-open arcs, each centered at $c_k + \tau/H_0$ and having length Δ .

Proposition 3.1 (Exact residue-comb partition). *The cells satisfy*

$$\mathbb{T} = \bigsqcup_{k \bmod L} \mathcal{C}_k \quad a.e., \quad |\mathcal{C}_k| = \frac{1}{L}. \quad (32)$$

The operators

$$\Pi_k := I_{\mathcal{V}} \otimes \mathcal{F}_j^{-1} M_{\mathbf{1}_{\mathcal{C}_k}} \mathcal{F}_j \quad (33)$$

are orthogonal projections and give the strong resolution

$$\Pi_k \Pi_\ell = \mathbf{1}_{k=\ell} \Pi_k, \quad \sum_{k \bmod L} \Pi_k = I_{\tilde{\mathcal{H}}}. \quad (34)$$

They commute with every bounded operator acting only on $\mathcal{V}$.

Proof. The intervals I_k partition the quotient circle and all have measure $1/L$. Pullback under π preserves normalized Haar measure. Multiplication by their indicators gives mutually orthogonal projections summing to the identity on $L^2(\mathbb{T})$. Conjugate by $\mathcal{F}_j$ and tensor with $I_{\mathcal{V}}$. The commutation statement follows because the operators act on different tensor factors. $\square$

The half-open convention affects only null boundaries. Centered closed arcs are convenient for support statements, but would overlap at their endpoints and should not be used to state an exact partition.

3.2 Physical-window trace

Let $I \subset \mathbb{Z}$ contain N orbit times, fix a unit vector $v \in \mathcal{V}$, and let

$$E_{I,v} : \ell^2(I) \longrightarrow \tilde{\mathcal{H}}, \quad E_{I,v} a := v \otimes \tilde{a}, \quad (35)$$

where $\tilde{a}$ is zero extension to $\ell^2(\mathbb{Z})$. This isolates one scalar non-orbit fiber and avoids taking a trace over $\mathcal{V}$.

Theorem 3.2 (Exact per-band trace). *On this scalar non-orbit fiber,*

$$\boxed{\operatorname{tr}(E_{I,v}^* \Pi_k E_{I,v}) = \|\Pi_k E_{I,v}\|_{\text{HS}}^2 = \frac{N}{L}.} \quad (36)$$

If only the progression $j \equiv a \pmod{H_0}$ is retained, the trace is $|I \cap (a + H_0\mathbb{Z})|/L$. Summing over k recovers the dimension of the corresponding finite orbit window.

Proof. The diagonal Fourier matrix coefficient of Π_k is $|\mathcal{C}_k| = 1/L$. Summation over the selected orbit coordinates proves the trace formula. Orthogonality gives the Hilbert–Schmidt identity. $\square$

At the inherited scale $N \asymp J$, the critical choice

$$L \asymp J \quad (37)$$

has bounded trace per full scalar orbit fiber. The individual compressed operators remain positive contractions rather than finite window projections, but the enlarged-space resolution (34) is exact.

3.3 Conservation under residual grouping

Let C be any bounded operator on the non-orbit factor; in the physical application it is the residual grouping map of TPC–47. Since $C\Pi_k = \Pi_k C$, Parseval gives

$$\sum_{k \bmod L} \|C\Pi_k y\|^2 = \|Cy\|^2 \leq \|C\|^2 \|y\|^2. \quad (38)$$

On the active residual space of TPC–47,

$$\|C\|^2 = \rho_{\text{res}}, \quad \rho_{\text{res}} = \max_{\gamma} |\mathcal{S}_{\gamma}|. \quad (39)$$

Thus all-band tiling introduces no additional grouping loss, but it does not bound ρ_{res} by $X^{o(1)}$.

4 Determinant resonance channels

Throughout this section $L \geq 2$, as in the critical application, and all determinant slopes m are positive integers.

For $t \in \mathbb{T}$, let

$$(U_t y)_j := e(jt) y_j. \quad (40)$$

Then U_t is unitary and

$$\mathcal{F}_j(U_t y)(\beta) = \widehat{y}(\beta - t). \quad (41)$$

The determinant equation contributes $t = m\alpha$.

4.1 The input-output band matrix

For $m \geq 1$, $\alpha \in \mathbb{T}$, and $q \in \mathbb{Z}/L\mathbb{Z}$, define

$$R_{m,q}(\alpha) := \sum_{\ell \bmod L} \Pi_{\ell+q} U_{m\alpha} \Pi_{\ell}. \quad (42)$$

All indices in this section are modulo L .

Theorem 4.1 (Exact resonance-channel resolution). *For every m, α ,*

$$U_{m\alpha} = \sum_q R_{m,q}(\alpha), \quad \sum_q \|R_{m,q}(\alpha)y\|^2 = \|y\|^2. \quad (43)$$

Moreover,

$$R_{m,q}(\alpha) \neq 0 \iff d_G(m\alpha, c_q) < \Delta, \quad (44)$$

where $d_G(x, y) = \text{dist}(x - y, G)$. For almost every α , exactly two values of q satisfy (44).

Proof. Insert the two resolutions of the identity around $U_{m\alpha}$:

$$U_{m\alpha} = \sum_{k,\ell} \Pi_k U_{m\alpha} \Pi_\ell,$$

and group the pair by $q = k - \ell$. In orbit Fourier variables, the (k, ℓ) -block is multiplication by

$$\mathbf{1}_{\mathcal{C}_k}(\beta) \mathbf{1}_{\mathcal{C}_\ell}(\beta - m\alpha), \quad (45)$$

followed by translation. For fixed m, α, β , there is one output cell k containing β and one input cell ℓ containing $\beta - m\alpha$. Hence exactly one $q = k - \ell$ contributes pointwise. This proves both identities in (43).

The shifted input interval and the output interval have equal length $1/L$ in the quotient coordinate. Their intersection has positive measure precisely when their centers are separated by less than $1/L$, which is (44) after division by H_0 . A point on a circle partitioned by centers spaced $1/L$ lies within distance $1/L$ of exactly two centers, except at the finitely many center points. This proves the last statement. $\square$

Define the resonance tube

$$\mathfrak{T}_{m,q}^{(L,H_0)} := \{\alpha \in \mathbb{T} : d_G(m\alpha, c_q) < \Delta\}. \quad (46)$$

Corollary 4.2 (Tube measure and branch width). *For every m, q ,*

$$|\mathfrak{T}_{m,q}^{(L,H_0)}| = \frac{2}{L}. \quad (47)$$

It is the union of mH_0 intervals, each with half-width

$$\frac{\Delta}{m} = \frac{1}{LH_0m}. \quad (48)$$

For almost every α ,

$$\sum_{q \bmod L} \mathbf{1}_{\mathfrak{T}_{m,q}^{(L,H_0)}}(\alpha) = 2. \quad (49)$$

Proof. Multiplication by m preserves normalized Haar measure on $\mathbb{T}$. The target neighborhood is a union of H_0 arcs of total measure $2H_0\Delta = 2/L$. Pulling those arcs back gives the branch count and half-width. The final identity is the last assertion of [theorem 4.1](#). $\square$

At $L \asymp J$ and $m \asymp M$,

$$\frac{1}{LH_0m} = X^{-1+o(1)}. \quad (50)$$

No band summation is involved in this scale: the two active channels already give the exact resolution of one determinant modulation.

4.2 The optimal channel-lift constant

Let $\mathcal{M} \subset \mathbb{N}$ be a finite set of distinct positive slopes and put $K = |\mathcal{M}|$. For $\mathbf{y} = (y_m)_{m \in \mathcal{M}} \in \bigoplus_m \tilde{\mathcal{H}}$, define

$$(\mathcal{A}_{\alpha,L}\mathbf{y})_q := \sum_{m \in \mathcal{M}} R_{m,q}(\alpha) y_m \in \tilde{\mathcal{H}}. \quad (51)$$

For almost every β , let $k(\beta)$ be the unique cell index with $\beta \in \mathcal{C}_{k(\beta)}$, and define

$$n_{\alpha,L}(\beta, q) := \#\{m \in \mathcal{M} : \beta - m\alpha \in \mathcal{C}_{k(\beta)-q}\}, \quad (52)$$

$$R_L(\alpha; \mathcal{M}) := \max_{q \bmod L} \operatorname{ess\,sup}_{\beta \in \mathbb{T}} n_{\alpha,L}(\beta, q). \quad (53)$$

Theorem 4.3 (Exact channel-lift norm). *The optimal Hilbert-space constant is*

$$\boxed{\|\mathcal{A}_{\alpha,L}\|^2 = R_L(\alpha; \mathcal{M})}. \quad (54)$$

In particular,

$$\sum_q \left\| \sum_m R_{m,q}(\alpha) y_m \right\|^2 \leq R_L(\alpha; \mathcal{M}) \sum_m \|y_m\|^2. \quad (55)$$

Proof. After orbit Fourier transform, the q -output at β is the sum of exactly the input fibers counted by $n_{\alpha,L}(\beta, q)$. Pointwise Cauchy–Schwarz, followed by integration and the one-channel-per-input identity in [theorem 4.1](#), gives the upper bound.

For the reverse inequality, choose q and a positive-measure set E on which the same set $\mathcal{M}_0$ of $R_L(\alpha; \mathcal{M})$ slopes is counted. Such a set exists because only finitely many subsets occur. Fix a unit vector $v \in \mathcal{V}$, and for $m \in \mathcal{M}_0$ set

$$\hat{y}_m(\gamma) = \mathbf{1}_{E-m\alpha}(\gamma)v;$$

set all other inputs to zero. Every selected modulation produces $\mathbf{1}_E v$ in the same q -channel. The squared output/input norm ratio is therefore $|\mathcal{M}_0| = R_L(\alpha; \mathcal{M})$. $\square$

For every fixed β , the K slopes are distributed among L values of q . Hence

$$R_L(\alpha; \mathcal{M}) \geq \left\lceil \frac{K}{L} \right\rceil. \quad (56)$$

This lower bound is geometric and makes no assertion about which input vectors occur in the actual TPC coefficient.

5 All-band conservation and the reassembly barrier

The output-side square function is exact and useful bookkeeping. It must not be confused with a new atomic estimate.

5.1 Amplitude and diagonal conservation

Let $Sx = \sum_{e \in \mathcal{E}} x_e u_e$ be any finite synthesis in $\tilde{\mathcal{H}}$, and define

$$\mathcal{D}(x) := \sum_{e \in \mathcal{E}} |x_e|^2 \|u_e\|^2. \quad (57)$$

For each orbit band, put

$$S_k x := \Pi_k Sx, \quad \mathcal{D}_k(x) := \sum_e |x_e|^2 \|\Pi_k u_e\|^2. \quad (58)$$

Proposition 5.1 (Exact all-band conservation). *For every x ,*

$$\sum_k \|S_k x\|^2 = \|Sx\|^2, \quad \sum_k \mathcal{D}_k(x) = \mathcal{D}(x). \quad (59)$$

Consequently, if $\|S_k x\|^2 \leq C \mathcal{D}_k(x)$ for every k with one common constant C , then $\|Sx\|^2 \leq C \mathcal{D}(x)$. Conversely, whenever $\mathcal{D}(x) > 0$, at least one band with $\mathcal{D}_k(x) > 0$ satisfies

$$\frac{\|S_k x\|^2}{\mathcal{D}_k(x)} \geq \frac{\|Sx\|^2}{\mathcal{D}(x)}. \quad (60)$$

Proof. The first identity is Parseval for the orthogonal resolution (34). Apply the same identity to each atom u_e , multiply by $|x_e|^2$, and sum to obtain the diagonal identity. Summing the bandwise inequalities gives the first consequence. The full ratio is the $\mathcal{D}_k$ -weighted average of the nonzero band ratios, which proves the second. $\square$

Thus an output tiling cannot hide a pre-existing coherent-box or tensor-to-atomic obstruction: some band inherits at least the full ratio. The conclusion is abstract and does not say that every actual band is bad.

5.2 Collapse of the difference channels

Let

$$\mathcal{Q}_L : \bigoplus_{q \bmod L} \tilde{\mathcal{H}} \longrightarrow \tilde{\mathcal{H}}, \quad \mathcal{Q}_L((z_q)_q) := \sum_q z_q. \quad (61)$$

Then $\|\mathcal{Q}_L\|^2 = L$. The channel lift factors the unlabelled determinant synthesis:

$$\mathcal{Q}_L \mathcal{A}_{\alpha, L} \mathbf{y} = \sum_{m \in \mathcal{M}} U_{m\alpha} y_m =: T_\alpha \mathbf{y}. \quad (62)$$

Theorem 5.2 (Sharp unrestricted reassembly barrier). *For every $\alpha \in \mathbb{T}$,*

$$\boxed{\|T_\alpha\|^2 = K}. \quad (63)$$

Hence

$$K \leq LR_L(\alpha; \mathcal{M}), \quad (64)$$

and no universal inequality for arbitrary independent orbit inputs can replace the factor K by a smaller all-band constant.

Proof. Cauchy–Schwarz gives $\|\sum_m U_{m\alpha} y_m\|^2 \leq K \sum_m \|y_m\|^2$. For a nonzero finitely supported g , take $y_m = U_{m\alpha}^{-1} g$. Then the output is Kg , while the input norm squared is $K\|g\|^2$, proving equality. The product bound is also immediate from $T_\alpha = \mathcal{Q}_L \mathcal{A}_{\alpha, L}$ and [theorem 4.3](#). $\square$

The witness is allowed in the ambient arbitrary-input class and can be finitely supported in the orbit coordinate. It is not asserted to be an actual source-mask coefficient. Its role is to identify exactly which extra structure a positive continuation must use.

5.3 A no-wrap collision bound

On the shift range relevant below, the slope points do not wrap around the quotient circle.

Proposition 5.3 (Elementary tube-load bounds). *Assume $\mathcal{M} \subset [M_-, M_+] \cap \mathbb{Z}$, $0 < t < 1$, $L \leq M_+$, and*

$$H_0 M_+ t < 1. \quad (65)$$

Then

$$\left\lceil \frac{K}{L} \right\rceil \leq R_L(t \pmod{1}; \mathcal{M}) \leq \min \left\{ K, 1 + \frac{2}{LH_0 t} \right\}. \quad (66)$$

If, more explicitly,

$$\mathcal{M} \subset [M, 2M] \cap \mathbb{Z}, \quad K \asymp M, \quad L \leq M, \quad (67)$$

and

$$\frac{a}{M} \leq t \leq \frac{b}{M}, \quad 0 < a < b < \frac{1}{2H_0}, \quad (68)$$

then

$$R_L(t \pmod{1}; \mathcal{M}) \asymp_{a,b,H_0} \frac{M}{L}. \quad (69)$$

Proof. The lower bound is (56). Under (65), the points $H_0 m t$ are separated by at least $H_0 t$ on an interval of length less than one. A quotient cell has length $1/L$, so it contains at most $1 + 2(LH_0 t)^{-1}$ such points; the factor two harmlessly covers a cell meeting the chosen cut of the quotient circle. This is the upper bound. On (68), the upper bound is $O(M/L)$, while the dense-slope hypothesis and the pigeonhole lower bound give the matching estimate. $\square$

For the critical choice $L \asymp J$, a dense slope box therefore has channel load M/J on the coarse-edge slab. However, multiplying by the L -channel collapse returns M , exactly as (64) predicts.

6 A sharp large sieve for a common orbit profile

The barrier in theorem 5.2 permits each slope to choose an independent compensating orbit vector. The projective layers in (27) do not have that freedom: all slopes share the same sampled scalar profile. This section quantifies the resulting gain.

6.1 The progression Gram kernel

Fix $a \pmod{H_0}$, $J \geq 1$, $\phi \in C_c^\infty(\mathbb{R})$, and a finite slope set $\mathcal{M} \subset [M_-, M_+] \cap \mathbb{Z}$. For Hilbert vectors $b_m \in \mathcal{V}$, define

$$\mathcal{L}_{\phi,a,\alpha}(\mathbf{b}) := \sum_{\substack{j \in \mathbb{Z} \\ j \equiv a \pmod{H_0}}} \phi(j/J) \left(\sum_{m \in \mathcal{M}} b_m e(mj\alpha) \right) \otimes e_j. \quad (70)$$

Put $K = |\mathcal{M}|$, $t = \|\alpha\|_{\mathbb{T}}$, and

$$D_J(t; K) := \begin{cases} \min \left\{ K, 1 + \frac{1}{Jt} \right\}, & t > 0, \\ K, & t = 0. \end{cases} \quad (71)$$

Let $\psi = |\phi|^2$. The Gram kernel is

$$G_d(\alpha) := \sum_{j \equiv a \pmod{H_0}} \psi(j/J) e(dj\alpha), \quad d \in \mathbb{Z}. \quad (72)$$

Lemma 6.1 (No-wrap progression decay). *Assume $t \geq 0$ and*

$$H_0(M_+ - M_-)t \leq \frac{1}{4}. \quad (73)$$

For every integer $B \geq 2$,

$$|G_d(\alpha)| \leq C_{B,H_0,\phi} \frac{J}{H_0} (1 + J|d|t)^{-B} \quad (74)$$

uniformly for $|d| \leq M_+ - M_-$.

Proof. Progression Poisson summation gives

$$G_d(\alpha) = \frac{J}{H_0} \sum_{q \in \mathbb{Z}} e(aq/H_0) \hat{\psi} \left(\frac{J}{H_0} (q - H_0 d \alpha) \right). \quad (75)$$

Choose the representative of α with absolute value t . Under (73), the $q = 0$ term is bounded by the right side of (74). Every $q \neq 0$ satisfies $|q - H_0 d \alpha| \geq 3/4$. Apply Schwartz decay at order $B + 2$ to those terms. If $J \geq H_0$, their sum after the prefactor J/H_0 is $O_{B,H_0,\phi}(J^{-B-1})$, which is bounded by the right side of (74) because $1 + J|d|t \leq 1 + J/(4H_0)$. If $J < H_0$, the conclusion is absorbed into the constant. Thus only a fixed finite Schwartz seminorm of ψ is used. $\square$

6.2 The Hilbert-valued inequality

Theorem 6.2 (Common-profile determinant large sieve). *Under (73),*

$$\|\mathcal{L}_{\phi,a,\alpha}(\mathbf{b})\|^2 \leq C_{\phi,H_0} \frac{J}{H_0} D_J(t; K) \sum_{m \in \mathcal{M}} \|b_m\|^2. \quad (76)$$

The constant is uniform in the Hilbert space, the vectors b_m , the slope locations inside $[M_-, M_+]$, and J .

Proof. If $t = 0$, Cauchy–Schwarz in the m -sum and $\sum_{j \equiv a \pmod{H_0}} |\phi(j/J)|^2 \ll_{\phi,H_0} J/H_0$ give (76) with $D_J(0; K) = K$. Assume henceforth that $t > 0$. Expanding the square in (70) gives the Hilbert-valued Gram form

$$\sum_{m,n \in \mathcal{M}} G_{m-n}(\alpha) \langle b_m, b_n \rangle. \quad (77)$$

By lemma 6.1 with $B = 2$,

$$\sup_m \sum_{n \in \mathcal{M}} |G_{m-n}(\alpha)| \ll_{\phi,H_0} \frac{J}{H_0} \left(1 + \frac{1}{Jt} \right). \quad (78)$$

The Schur test, valid after scalar pairing in $\mathcal{V}$, gives this constant. The trivial estimate $|G_d| \ll_{\phi,H_0} J/H_0$ gives instead $(J/H_0)K$. Taking the smaller bound proves (76). $\square$

By orbit Plancherel, the left side is also the complete all-band square function

$$\|\mathcal{L}_{\phi,a,\alpha}(\mathbf{b})\|^2 = \sum_{k \bmod L} \|\Pi_k \mathcal{L}_{\phi,a,\alpha}(\mathbf{b})\|^2 \quad (79)$$

for every choice of L . In particular, the estimate already controls the reassembled orbit field and carries no factor equal to the number of bands.

6.3 Sharpness in the common-profile class

Proposition 6.3 (Clustered-slope lower bound). *Assume $t > 0$, $\phi \neq 0$, and $J \geq J_0(\phi, H_0)$. There are constants $c_{\phi, H_0}, C_{\phi, H_0} > 0$ with the following property. Suppose $\mathcal{M}$ contains D consecutive integers and either $D = 1$, or*

$$2 \leq D \leq c_{\phi, H_0} \min\{K, (Jt)^{-1}\}. \quad (80)$$

Then one can choose collinear Hilbert vectors, supported on those slopes, such that

$$\|\mathcal{L}_{\phi, a, \alpha}(\mathbf{b})\|^2 \geq C_{\phi, H_0}^{-1} \frac{J}{H_0} D \sum_m \|b_m\|^2. \quad (81)$$

Consequently $D_J(t; K)$ is optimal up to constants for dense common-profile slope boxes.

Proof. Choose an interval on which $|\phi|$ is bounded below. The lower bound on J ensures that its J -dilate contains $\gg_{\phi, H_0} J/H_0$ points in the progression, uniformly in the finitely many residue classes a . For $D = 1$, any one supported slope gives the claim directly. For $D \geq 2$, let j_0 be the center of the sampled interval and choose the signed lift $\tilde{\alpha} \in [-1/2, 1/2]$ with $|\tilde{\alpha}| = t$. Fix a unit vector v , and on a consecutive block put $b_m = e(-mj_0\tilde{\alpha})v$. For j in a sufficiently short fixed subinterval around j_0 , condition (80) makes all phases $e(m(j - j_0)\tilde{\alpha})$ lie in one fixed acute arc. Their sum has modulus $\gg D$. Summing over those j gives $\gg_{\phi, H_0} (J/H_0)D^2$, whereas $\sum_m \|b_m\|^2 = D$. $\square$

Taking $D = 1$ when $Jt \gg 1$, and $D \asymp_{\phi, H_0} \min\{K, (Jt)^{-1}\}$ when $Jt \ll 1$, proves the stated optimality for slope boxes containing the required consecutive cluster. At $t = 0$, choosing all K vectors equal gives the exact factor K . The sharpness witness uses arbitrary collinear coefficients inside one admissible common-profile class. It does not assert that the literal actual coefficients realize this coherence.

7 Reduction of the complete frozen actual-coefficient family

We now apply [theorem 6.2](#) to the full signed representation (26). No orbit band and no projective layer is selected.

7.1 The row-square normal form

For one projective layer, define

$$B_{\omega, m}(\alpha) := \sum_{p, r} A_{\omega}(p, r, m) e(-pr\alpha) v_{\omega, p, r, m} \in \mathcal{V}_L. \quad (82)$$

The arithmetic sum is finite. Fourier transformation in the frozen determinant intercept gives the exact identity

$$\widehat{\mathcal{F}}^{\omega}(\alpha) = \sum_{\substack{j \in \mathbb{Z} \\ j \equiv a_{\omega} \pmod{H_0}}} \phi_{\omega}(j/J) \left(\sum_m B_{\omega, m}(\alpha) e(mj\alpha) \right) \otimes e_j. \quad (83)$$

Proposition 7.1 (Exact residual square at fixed source). *For fixed ω, m, α , put*

$$B_{\omega, m, r}(\alpha) := \sum_p A_{\omega}(p, r, m) e(-pr\alpha) v_{\omega, p, r, m}. \quad (84)$$

Then

$$\boxed{\|B_{\omega, m}(\alpha)\|^2 = \sum_r \|B_{\omega, m, r}(\alpha)\|^2.} \quad (85)$$

Proof. For fixed m , the vector $B_{\omega,m,r}(\alpha)$ is supported on the residual coordinate $s = d(m)r$. Distinct supported cofactors r give distinct labels and hence orthogonal coordinate subspaces. The prime sum at one fixed r remains coherent, which is why no p -square decomposition is claimed. $\square$

This identity localizes the remaining arithmetic task: one must control the prime exponential sum inside each residual coordinate and then the outer square sum over m, r .

7.2 A representation-independent projective envelope

Let $C_{\text{LS}}(\phi, H_0)$ denote the least admissible constant C_{ϕ, H_0} in (76). For an admissible signed representation $\mathcal{R}$ of the actual coefficient, set

$$\mathfrak{S}_{\mathcal{R}}(\alpha) := \int C_{\text{LS}}(\phi_{\omega}, H_0)^{1/2} \left(\sum_m \|B_{\omega,m}(\alpha)\|^2 \right)^{1/2} d|\nu|(\omega), \quad (86)$$

and define

$$\mathfrak{S}_{\text{pr}}(\alpha) := \inf_{\mathcal{R}} \mathfrak{S}_{\mathcal{R}}(\alpha), \quad (87)$$

where the infimum ranges over all inherited admissible coefficientwise representations described in section 2. The set is nonempty by TPC–36/47. This definition does not estimate the envelope; it makes the reduction independent of one chosen representation. It is used only pointwise. For integration we keep one fixed global representation, thereby avoiding any measurability assertion for an infimum over an uncountable representation class.

The proof of theorem 6.2 shows that $C_{\text{LS}}(\phi, H_0)^{1/2}$ is controlled by one fixed finite-order C^B -seminorm once the common compact support is fixed. For every fixed admissible representation, the layer data are jointly measurable and Bochner integrable; consequently $\mathfrak{S}_{\mathcal{R}}$ is a measurable extended-valued function. Finiteness almost everywhere is part of admissibility. If one works with a broader representation class, all pointwise bounds remain valid with value $+\infty$, while the annular application below explicitly requires a finite envelope.

Theorem 7.2 (All-band reduction of the actual frozen family). *Let $t = \|\alpha\|_{\mathbb{T}} \geq 0$, and assume (73). Then the complete left frozen family built from the literal h_0 -coefficient field satisfies*

$$\|\widehat{\mathcal{F}}^{\text{fr},L}(\alpha)\| \leq \left(\frac{J}{H_0} D_J(t; K) \right)^{1/2} \mathfrak{S}_{\text{pr}}(\alpha). \quad (88)$$

The same argument gives the right-side estimate with its corresponding support cardinality and envelope. Denote the preceding left envelope by $\mathfrak{S}_{\text{pr},L}$ and its right analogue by $\mathfrak{S}_{\text{pr},R}$. On the orthogonal left–right sum, the valid combined envelope is $(\mathfrak{S}_{\text{pr},L}^2 + \mathfrak{S}_{\text{pr},R}^2)^{1/2}$, with the larger of the two large-sieve prefactors if their supports differ. Equivalently, for every band count L ,

$$\left(\sum_{k \bmod L} \|\Pi_k \widehat{\mathcal{F}}^{\text{fr},L}(\alpha)\|^2 \right)^{1/2} \leq \left(\frac{J}{H_0} D_J(t; K) \right)^{1/2} \mathfrak{S}_{\text{pr}}(\alpha). \quad (89)$$

Proof. Apply theorem 6.2 to (83). Minkowski’s integral inequality and the exact signed identity (26) give (88) with $\mathfrak{S}_{\mathcal{R}}$ for every admissible representation. Take the infimum. Equation (89) is the exact orthogonal resolution (34). $\square$

The theorem controls the complete orbit output, not a selected band, and it incurs no cover-count factor. Its right side is a projective row-square envelope rather than the physical atomic diagonal. No bound relating those two quantities is assumed.

7.3 Residual grouping

On the active residual space, let C be the grouping map from TPC-47. It acts on the non-orbit factor, commutes with every Π_k and $U_{m\alpha}$, and satisfies $\|C\|^2 = \rho_{\text{res}}$. Consequently

$$\|C\widehat{\mathcal{F}}^{\text{fr},L}(\alpha)\| \leq \rho_{\text{res}}^{1/2} \left(\frac{J}{H_0} D_J(t; K) \right)^{1/2} \mathfrak{S}_{\text{pr}}(\alpha). \quad (90)$$

At h_0 , this grouping recovers the corresponding large-prime component of the actual equality column under the inherited global sign convention. It neither restores the base component nor converts the residual Gram route into the native affine physical face.

8 Endpoint ledger and the annular resonance budget

We now insert the inherited TPC endpoint. Write

$$\mu := \frac{267}{400}, \quad \nu := \frac{133}{400}, \quad \kappa := \frac{1}{400}. \quad (91)$$

Thus

$$M = X^{\mu+o(1)}, \quad J = X^{\nu+o(1)}, \quad \mu + \nu = 1. \quad (92)$$

The actual slope support only obeys $K \leq X^{\mu+o(1)}$. Statements that identify K with M below are therefore model sharpness statements for a dense slope box, not lower bounds for the literal coefficient.

8.1 Critical bands and exact fine tubes

Take

$$L = X^{\nu+o(1)} \asymp J. \quad (93)$$

For a full scalar orbit window of length $N \asymp J$, [theorem 3.2](#) gives

$$\text{tr}(E_{I,v}^* \Pi_k E_{I,v}) = O(1). \quad (94)$$

For $m \asymp M$, [corollary 4.2](#) gives the exact branch half-width

$$\frac{1}{LH_0m} = X^{-1+o(1)}. \quad (95)$$

The fine scale is therefore recovered simultaneously across an exact orthogonal resolution of the whole orbit-frequency circle. This is a geometric resolution statement; it is not a mass lower bound or a spectral gap for the complete actual vector.

It is useful to compare this choice with the canonical band in TPC-47. There the band count needed to cover the quotient was

$$L_{47} = X^{\nu-\kappa+o(1)} = X^{33/100+o(1)}, \quad (96)$$

the per-band trace was $X^{\kappa+o(1)}$, and the resonance half-width was $X^{-1+\kappa+o(1)} = X^{-399/400+o(1)}$. Passing to the critical tiling adds the missing factor X^κ to the band count, makes the trace bounded, and sharpens the tube scale by the same factor. Unlike the single projected sector in TPC-47, the present L bands sum exactly to the identity.

8.2 Pointwise collision ledger

On the annular shift range below, the no-wrap hypothesis holds and the large-sieve factor is

$$D_J(t; K) = \min \left\{ K, 1 + \frac{1}{Jt} \right\}. \quad (97)$$

At the fine edge $t = X^{-1+\epsilon}$, the universal upper bound $K \leq X^{\mu+o(1)}$ gives

$$D_J(t; K) \leq X^{\mu-\epsilon+o(1)}, \quad \frac{J}{H_0} D_J(t; K) \leq X^{1-\epsilon+o(1)}. \quad (98)$$

At the coarse edge $t = X^{-\mu-\epsilon}$,

$$D_J(t; K) \leq X^{\mu-\nu+\epsilon+o(1)} = X^{67/200+\epsilon+o(1)}, \quad (99)$$

and hence

$$\frac{J}{H_0} D_J(t; K) \leq X^{\mu+\epsilon+o(1)}. \quad (100)$$

The corresponding amplitude prefactors are

shift scale	squared prefactor	amplitude prefactor
$t = X^{-1+\epsilon}$	$X^{1-\epsilon+o(1)}$	$X^{1/2-\epsilon/2+o(1)}$
$t = X^{-\mu-\epsilon}$	$X^{\mu+\epsilon+o(1)}$	$X^{\mu/2+\epsilon/2+o(1)}$

For a dense slope box at the coarse scale $t \asymp M^{-1}$, inside the no-wrap slab (68), the normalized collision load is

$$\frac{M}{J} = X^{\mu-\nu+o(1)} = X^{67/200+o(1)}. \quad (101)$$

The inherited TPC allowance is the normalized determinant-packet loss

$$K_B = X^{\kappa+o(1)} \quad (102)$$

from the TPC-45/47 closure ledger [5, 7]. The unresolved dense-model exponent gap is therefore

$$(\mu - \nu) - \kappa = \frac{67}{200} - \frac{1}{400} = \frac{133}{400} = \nu. \quad (103)$$

Thus the common-profile theorem saves the full orbit length relative to the unrestricted K -loss, but the coarse-edge bound still does not fit the atomic TPC budget. Attainment of this common-profile operator scale requires the consecutive slope cluster in [proposition 6.3](#); it is not asserted for the literal coefficient.

8.3 A logarithmic relative annulus theorem

For

$$0 < \epsilon < \frac{\nu}{2} = \frac{133}{800}, \quad \mathcal{A}_\epsilon := \left\{ \alpha \in \mathbb{T} : X^{-1+\epsilon} \leq \|\alpha\|_{\mathbb{T}} \leq X^{-\mu-\epsilon} \right\}, \quad (104)$$

the annulus is nonempty for large X . Moreover,

$$H_0(M_+ - M_-) \|\alpha\|_{\mathbb{T}} = o(1) \quad (105)$$

uniformly on $\mathcal{A}_\epsilon$, after the inherited fixed dyadic localization. Hence [theorem 7.2](#) applies there.

Fix one global admissible signed representation $\mathscr{R}$. It is important here not to integrate the pointwise infimum $\mathfrak{S}_{\text{pr}}$, for which no measurability assertion was made.

Theorem 8.1 (Relative annular resonance budget). *For every fixed global admissible representation $\mathcal{R}$, with the measurable envelope specified in (86),*

$$\int_{\mathcal{A}_\epsilon} \|\widehat{\mathcal{F}}^{\text{fr},\text{L}}(\alpha)\|^2 d\alpha \leq \frac{J}{H_0} \int_{\mathcal{A}_\epsilon} D_J(\|\alpha\|_{\mathbb{T}}; K) \mathfrak{S}_{\mathcal{R}}(\alpha)^2 d\alpha. \quad (106)$$

In particular, if

$$\sup_{\alpha \in \mathcal{A}_\epsilon} \mathfrak{S}_{\mathcal{R}}(\alpha) \leq S_*, \quad (107)$$

then

$$\int_{\mathcal{A}_\epsilon} \|\widehat{\mathcal{F}}^{\text{fr},\text{L}}(\alpha)\|^2 d\alpha \ll_{\epsilon, H_0} (\log X) S_*^2. \quad (108)$$

The contribution of this annulus to Fourier inversion at h_0 satisfies

$$\left\| \int_{\mathcal{A}_\epsilon} \widehat{\mathcal{F}}^{\text{fr},\text{L}}(\alpha) e(h_0 \alpha) d\alpha \right\| \ll X^{-(\mu+\epsilon)/2+o(1)} (\log X)^{1/2} S_*. \quad (109)$$

Proof. The first estimate is the squared version of (88), retained for the fixed representation $\mathcal{R}$. The elementary inequality

$$D_J(t; K) \leq 1 + \frac{1}{Jt} \quad (110)$$

holds throughout the annulus. By symmetry around zero,

$$\int_{\mathcal{A}_\epsilon} D_J(\|\alpha\|_{\mathbb{T}}; K) d\alpha \ll X^{-\mu-\epsilon} + \frac{1}{J} \log \frac{X^{-\mu-\epsilon}}{X^{-1+\epsilon}} \quad (111)$$

$$\ll X^{-\nu+o(1)}. \quad (112)$$

Multiplication by J/H_0 proves (108). Finally $|\mathcal{A}_\epsilon| \ll X^{-\mu-\epsilon}$, so Cauchy–Schwarz in α gives (109). $\square$

The theorem is deliberately relative to S_* . Neither (108) nor (109) estimates the projective envelope by the physical atomic diagonal. It also leaves the ultra-low interval, the complementary shift frequencies, and any sparse alias reconstruction untreated.

9 Tube statistics and sharpness ledgers

The pointwise channel load of [theorem 4.3](#) is a worst-case quantity. A separate two-dimensional calculation records the exact average incidence geometry. It is useful precisely because it also shows what averaging does *not* prove.

Let $E \subset \mathbb{T}$ be measurable and let $\mathcal{M} \subset \mathbb{Z}$ contain K distinct integers. Define

$$\mathcal{L}_E(\alpha, \beta) := \sum_{m \in \mathcal{M}} \mathbf{1}_E(\beta - m\alpha). \quad (113)$$

Theorem 9.1 (Exact first and second tube moments). *With normalized Haar measure on $\mathbb{T}^2$,*

$$\int_{\mathbb{T}^2} \mathcal{L}_E(\alpha, \beta) d\alpha d\beta = K|E|, \quad (114)$$

$$\int_{\mathbb{T}^2} \mathcal{L}_E(\alpha, \beta)^2 d\alpha d\beta = K|E| + K(K-1)|E|^2. \quad (115)$$

Proof. Translation invariance gives $\int_{\mathbb{T}} \mathbf{1}_E(\beta - m\alpha) d\beta = |E|$, proving the first identity after summing over m . For the second, expand the square. The K diagonal terms contribute $K|E|$. If $m \neq n$, the map

$$(\alpha, \beta) \mapsto (\beta - m\alpha, \beta - n\alpha) \quad (116)$$

is a surjective continuous homomorphism of compact tori and therefore pushes normalized Haar measure to normalized Haar measure. Each ordered off-diagonal pair consequently contributes $|E|^2$. $\square$

For one critical quotient cell $E = \mathcal{C}_k$, $|E| = 1/L$, so

$$\mathbb{E}_{\alpha, \beta} \mathcal{L}_E = \frac{K}{L}, \quad \mathbb{E}_{\alpha, \beta} \mathcal{L}_E^2 = \frac{K}{L} + \frac{K(K-1)}{L^2}. \quad (117)$$

For dense $K \asymp M$ and critical $L \asymp J$, the mean load is

$$\frac{K}{L} \asymp \frac{M}{J} = X^{67/200+o(1)}. \quad (118)$$

This records the same K/L geometric density scale as the coarse-edge model in (69). It is not a fixed- α estimate.

Remark 9.2 (What the moment identity does not average). The variable β in (113) is free. The theorem does not give a second moment for the one-variable channel count at a fixed orbit frequency, for $R_L(\alpha; \mathcal{M})$, or for the literal source-mask coefficient. Nor may one insert an additional independent restriction $\beta \in \mathcal{C}_q$ and retain the same formula without a new calculation. The exact two-dimensional moments are an incidence ledger, not an arithmetic cancellation theorem.

9.1 The conserved phase-space product

TPC-47 used $L_{47} = X^{\nu-\kappa+o(1)}$ bands at its mesoscopic resolution. The corresponding dense coarse-edge load was

$$\frac{M}{L_{47}} = X^{\mu-\nu+\kappa+o(1)} = X^{27/80+o(1)}, \quad (119)$$

and the product satisfied

$$L_{47} \frac{M}{L_{47}} = M. \quad (120)$$

At the present critical resolution,

$$L \frac{M}{L} = M \quad (121)$$

as well. The additional factor X^κ in L removes the mesoscopic κ -loss from the resolved channel load and restores the exact X^{-1} tube scale, but it cannot remove the intrinsic $M/J = X^{\mu-\nu}$ density of source slopes per critical cell.

There are therefore two complementary sharpness statements:

- (i) arbitrary independent orbit inputs recover the full factor K after channel collapse, by [theorem 5.2](#);
- (ii) common-profile inputs have the smaller factor $D_J(t; K)$, and consecutive slope clusters attain it up to constants, by [proposition 6.3](#).

Each inherited projective layer lies inside the second class. The fully reassembled coefficient is instead a signed projective integral controlled through $\mathfrak{S}_{\mathcal{R}}$; no theorem in this paper says that it either saturates or improves the common-profile frontier.

10 What has changed, what remains, and the next gate

The present step closes the geometric all-band question raised in TPC–47. It does not close the arithmetic fixed-shift question. We record the distinction as a theorem ledger before describing a possible continuation.

10.1 Advances under the inherited exact interface

The following statements are proved here without a new arithmetic hypothesis beyond the exact TPC–36/45/47 representation and support interface.

- (i) The orbit-frequency quotient admits an exact orthogonal tiling by L full residue combs, with exact compressed trace N/L on every scalar physical window.
- (ii) Each determinant modulation has an exact input-output channel resolution. The support consists of explicit resonance tubes, and the critical branch half-width is $X^{-1+o(1)}$.
- (iii) The optimal norm of the channel lift is exactly the maximal pointwise tube multiplicity. Erasing the channel label restores the unrestricted synthesis norm squared K .
- (iv) The common orbit profile in every inherited projective layer yields the Hilbert-valued factor $D_J(t; K) = \min\{K, 1 + (Jt)^{-1}\}$, sharp up to constants in its declared class.
- (v) Complete signed reassembly, coefficient by coefficient, reduces the literal frozen family to a representation independent pointwise projective m -square envelope. At fixed m , residual labels give an exact outer r -square sum, while the prime sum inside each residual coordinate remains coherent.
- (vi) Relative to any fixed global projective representation with a bounded envelope, the intermediate annulus has only logarithmic integrated resonance cost.

These are forward results rather than a reformulation of the single-band theorem: all orbit bands are now controlled at once, the fine determinant scale is restored, and the exact obstruction left by their reassembly is identified.

10.2 The remaining arithmetic square-function gate

By (85), the unresolved layerwise object is

$$\sum_m \sum_r \left\| \sum_p A_\omega(p, r, m) e(-pr\alpha) v_{\omega, p, r, m} \right\|^2. \quad (122)$$

Three features prevent the present large sieve from estimating this quantity at the atomic scale:

- (a) the p -sum is coherent inside a fixed residual coordinate;
- (b) the projective integral must be reassembled with its literal signs and total variation, not by selecting a favorable layer;
- (c) subsequent residual grouping can cost $\rho_{\text{res}}^{1/2}$, for which no $X^{o(1)}$ bound has been inherited at this interface.

For the dense consecutive-slope sharpness model at the coarse edge, the common-profile collision factor exceeds the TPC allowance by the exponent $\nu = 133/400$. Any continuation that aims at atomic closure must therefore use arithmetic information inside (122); orbit geometry alone is now known to be insufficient.

10.3 Four precise doors

The next paper may attack any one of the following independently.

Door A: coefficient-specific prime square.

Prove cancellation in the coherent prime sum of (122), averaged over m, r , beyond the universal common-profile large sieve.

Door B: projective-to-atomic comparison.

Bound one fixed exact representation envelope $\mathfrak{S}_{\mathcal{R}}$ by the inherited physical atomic diagonal with a quantified loss, or construct an actual-mask witness showing that such a bound is impossible at the desired exponent.

Door C: residual grouping with cofactor structure.

Replace the abstract ρ_{res} loss by a coefficient-specific square-function estimate using the literal map $s = d(m)r$. The balanced cofactor fibers from earlier descent cannot be imported here without proving that the two interfaces coincide.

Door D: localization and alias completion.

Combine the annular theorem with an ultra-low analysis, an anti-alias or maximal-in-shift theorem, and the literal h_0 Fourier inversion. This door is downstream of, rather than a substitute for, the atomic comparison.

Each door has a meaningful negative outcome: an explicit actual-mask counterexample or a sharp lower bound for the necessary loss would locate the next obstruction just as the unrestricted synthesis theorem does here.

10.4 Prohibited conclusions

For clarity, none of the following is asserted.

- (i) No estimate of the complete physical atom by its atomic diagonal is proved.
- (ii) No bound $\rho_{\text{res}} = X^{o(1)}$, no new cofactor-range closure, and no cancellation of the internal prime sum is proved.
- (iii) No sparse-alias unfolding, new-shift physical provenance, or uniform local-shift theorem is proved. Only $h = h_0$ is the inherited physical residual synthesis; all other shifts are frozen algebraic extensions.
- (iv) The residual synthesis remains the low-sign-averaged Gram route and is not identified with the native affine physical face.
- (v) No parity-barrier breach, Hardy–Littlewood asymptotic, bounded prime-gap theorem, or twin-prime conclusion follows.

The exact conclusion is narrower and reusable: after full orbit-frequency resolution, the next non-geometric obstruction is the actual projective m, r -square of coherent prime sums. That is the next gate, not a disguised solution of it.

11 Deterministic finite-model certificate

The accompanying script

`experiments/tpc48_certificate.py`

uses only the Python standard library, exact finite-field arithmetic, integers, and the `Fraction` class. It uses no floating-point arithmetic, randomness, `assert` statements, or true division. The committed JSON record contains

$$68,662 \tag{123}$$

successful deterministic checks.

The principal finite model has

$$N = 60, \quad H_0 = 3, \quad L = J = 4, \quad K = 9. \tag{124}$$

It verifies an exact four-band Fourier partition, rank 15 in each band, compressed critical-window trace 1, the complete resonance tile/channel map, and tube first and second moment totals 540 and 1500. It also verifies that the unlabelled synthesis norm squared is $K = 9$, whereas the resolved channel collision norm squared in the chosen instance is $R = 4$. A separate rational Gram model checks a common-profile large-sieve inequality by an exact positive semidefinite LDL^* certificate. Finally, exact rational arithmetic checks all endpoint exponent identities in [section 8](#).

The normalized script hash is

`ead2a9584cd3680ce0c1f770dcec3058b3436ce5dfdecc1a0fcb3858cf067f64,`

the committed JSON SHA-256 hash is

`7e4ed4d83c5e00e441f3d8ee7aa1a6d9b00a37f975af2e673a743c0c83c72882,`

and the semantic certificate digest is

`da179eb730da85c38af54f73ebf2151aece5562db62bc5fb24e2bad81d77456e.`

The certificate is a regression witness for the exact algebra and for a finite cyclic analogue. It is not a proof of the continuous asymptotic large-sieve theorem, a bound for the actual projective envelope, a fixed-shift prime-pair estimate, or evidence that the parity barrier has been crossed.

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